Projective Geometry: Completing Lines and Preserving Incidence

N-20240518

Two parallel affine lines never meet in their familiar plane. The projective plane is built so that the apparent exception becomes an ordinary incidence statement. Its construction, and the geometry of cross-ratios and conics below, follow the standard sequence in the University of Oxford's projective geometry lectures.[1]

§1.1 Projective points are one-dimensional subspaces

Over a field k, define

$$\mathbb{P}^2(k) = (k^3 \setminus \{0\})/k^\times, \qquad [x : y : z] = [\lambda x : \lambda y : \lambda z] \quad (\lambda \neq 0).$$

A projective point represents an entire one-dimensional subspace of k^3; a projective line represents a two-dimensional subspace. Thus the projective line with equation

$$ax + by + cz = 0$$

is not merely an affine line extended very far away: it is the set of all one-dimensional subspaces contained in the indicated plane through the origin of k^3.

In the chart $z \neq 0$, rescale to $z = 1$ and obtain the affine coordinates $(x/z, y/z)$. The projective points with $z = 0$ constitute the line at infinity relative to *this chosen chart*.

§1.2 Where two parallel affine lines intersect

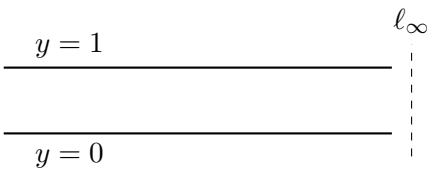
The affine equations $y = 0$ and $y = 1$ become the homogeneous projective equations

$$y = 0, \qquad y - z = 0.$$

Their intersection satisfies $y = z = 0$ and is therefore the single projective point

$$[1 : 0 : 0].$$

More generally, two distinct projective lines in $\mathbb{P}^2(k)$ arise from distinct two-dimensional subspaces of k^3; their intersection has dimension one. This linear-algebra argument proves that they meet in exactly one projective point.



The dashed line is a mnemonic for the missing directions, not a Euclidean line sitting at a large finite distance.

[1] https://courses.maths.ox.ac.uk/course/section.php?id=10755

§1.3 The dual plane and incidence

The coefficients $[a : b : c]$ of a line are themselves projective coordinates in the dual vector space $(k^3)^*$. The incidence relation

$$ax + by + cz = 0$$

pairs a point $[x : y : z]$ and a dual point $[a : b : c]$, and is independent of how either is scaled.

For distinct projective points u, v, a linear functional vanishing on their span describes their unique joining line. Dually, the two line functionals ℓ_1, ℓ_2 have a one-dimensional common kernel giving their unique intersection point. The dual formulations follow from the same rank-two linear algebra. No metric notions were needed.

§1.4 Projective Changes of Coordinates

An invertible matrix $A \in GL(3, k)$ acts by $[v] \mapsto [Av]$; scalar multiples of A give the same projective map, so the group is $PGL(3, k)$. As A takes subspaces to subspaces, it preserves incidence and collinearity.

It does not preserve Euclidean distances or angles. For example the affine shear

$$(x, y) \mapsto (x + y, y)$$

extends to a projective transformation but sends the perpendicular coordinate axes to nonperpendicular lines. Projective geometry studies properties surviving the larger group, not arbitrary measurements of a drawing.

§1.5 The cross-ratio really survives fractional linear maps

Write four distinct points on $\mathbb{P}^1(k)$ in an affine chart as a, b, c, d. Their cross-ratio is

$$[a, b; c, d] = \frac{(c - a)(d - b)}{(c - b)(d - a)}.$$

For $T(t) = (\alpha t + \beta)/(\gamma t + \delta)$ with $\alpha\delta - \beta\gamma \neq 0$, direct subtraction gives

$$T(u) - T(v) = \frac{(\alpha\delta - \beta\gamma)(u - v)}{(\gamma u + \delta)(\gamma v + \delta)}.$$

Substitute the four differences into the cross-ratio: every determinant factor and denominator cancels. Thus $[T(a), T(b); T(c), T(d)] = [a, b; c, d]$, wherever the affine expressions are defined. The homogeneous version extends this identity to points at infinity.

This calculation explains why cross-ratio, rather than length or angle, is a natural projective invariant.

§1.6 Conics and the choice of an affine window

A plane conic has a homogeneous quadratic equation $v^T Q v = 0$, with Q symmetric when the characteristic is not two. Under an invertible coordinate change $v = Au$, the same conic is represented by $u^T A^T Q A u = 0$. For nonsingular Q, this describes a nonsingular projective conic.

The familiar real distinctions among ellipse, parabola, and hyperbola arise from intersections with the chosen line at infinity. For instance, homogenizing

$$x^2 + y^2 = 1, \qquad y = x^2, \qquad x^2 - y^2 = 1$$

gives

$$x^2 + y^2 = z^2, \qquad x^2 = yz, \qquad x^2 - y^2 = z^2.$$

On $z = 0$, these have respectively no real points, one real point, and two real points (the point of tangency in the middle case has multiplicity two). A projective transformation can move the line at infinity, so these affine classifications need not survive it.

§1.7 A conic can be parametrized by a projective line

Consider the nonsingular conic $xz - y^2 = 0$. There is a map

$$\nu_2 : \mathbb{P}^1 \longrightarrow \mathbb{P}^2, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$

Its image lies on the conic because $s^2t^2 - (st)^2 = 0$. Conversely, if $x \neq 0$ on the conic, rescale to $[1 : y/x : (y/x)^2]$; the remaining point $[0 : 0 : 1]$ is the image of $[0 : 1]$. Thus ν_2 is an isomorphism onto the conic (over any field, with the usual scheme interpretation).

> **Caution**
>
> The conic is isomorphic to $\mathbb{P}^1$ as a projective algebraic curve; it cannot be mapped onto a projective *line in* $\mathbb{P}^2$ by a projective linear automorphism of $\mathbb{P}^2$. Such an automorphism preserves polynomial degree and sends lines to lines, whereas this embedding is quadratic.

The completion of affine lines, duality of points and lines, invariance of cross-ratio, and quadratic parametrization of a conic are different consequences of a single choice: use homogeneous coordinates and linear subspaces to record incidence without making a particular affine chart fundamental.